

Machine Learning Algorithms for Predicting and Preventive Diagnosis of Cardiovascular Disease

Gulbakyt Sembina
Information Systems Department
International Information Technology
University
Almaty, Kazakhstan
g.sembina@iitu.edu.kz

Aigerim Aitim
Information Systems Department
International Information Technology
University
Almaty, Kazakhstan
a.aitim@iitu.edu.kz

Medet Shaizat
Information Systems Department
International Information Technology
University
Almaty, Kazakhstan
24432@iitu.edu.kz

Abstract—The article shows the creation of 6 trained models for the diagnosis of cardiovascular diseases based on 6 different machine learning algorithms. The best machine learning algorithm is determined. The quality of the model was assessed according to such metrics as the proportion of correct answers, accuracy, and recall. An overview of already implemented projects for the use of artificial intelligence in medicine in Kazakhstan and in the world is presented. Using data of 224 patients, is created training dataset to further train the models.

Keywords—*medicine; cardiovascular disease; machine learning; artificial intelligence; machine learning algorithms; training dataset; classification; models*

I. INTRODUCTION

There is now a global shift from the treatment of consequences to a preventive health care system based on early diagnosis. Any medical form, examination protocol and anamnesis contain information that allows you to literally make a diagnosis and prescribe treatment. Unfortunately, even experienced doctors do not always see the perfect picture of the disease, since the data in the disease situation are not structured, and the disease situation can be very large. Their functionality is still affected by fatigue and, in some cases, a defect in knowledge in narrow areas. Some diseases, for example, oncological ones, can be won if non-obvious signs are distinguished in a timely manner and healing begins. The introduction of elements of artificial intelligence can solve this problem.

Machine learning is a field of artificial origin of the mind, which is increasingly being used in the field of cardiovascular medicine. In essence, for example, computers analyze the data and solve or classify the problem with or without the role of a person. Machine learning apron is based on models that take input and, through mathematical optimization and statistical analysis, predict outcomes. Several machine learning algorithms were used in everyday work. For example, a hackneyed machine learning method designated as SVM can learn non-linear patterns for use in face recognition, handwriting interpretation, or the detection of fraudulent credit card transactions. For example, called boosting

techniques used for prediction and systematization have been adapted to identify and process spam messages. Another method, called Random Forest (RF), could make it easier to make inferences by averaging multiple nodes. When processing a Convolutional Neural Network (CNN), it converges a few layers and apples to organize and segment images. We have previously laid out the technical details of each of these algorithms, but no consensus has been reached on the selection of specific algorithms for clinical use in the field of cardiovascular medicine. But choosing optimal algorithms for research questions and playing algorithms in all guises of clinical datasets is absolutely feasible, clinical interpretation and evaluation to implement the algorithms is quite difficult. Absolute awareness of the statistical and clinical knowledge of machine learning practitioners is still a challenge. Most ML studies have reported discrimination qualities such as area under the ROC curve (AUC) rather than p-values. Most importantly, it remains to be seen what the applicable cut-off value of AUC for use in medical practice, the interpretation of the cut-off value, and the appropriate/best practices for use in CVD datasets. Previously, we proposed a methodology for conducting machine learning studies in medicine. Regular educational program and meta-analysis, the basis of progressive evidence-based medicine, must be performed to evaluate the available method of machine [1].

II. PRACTICAL ASPECTS OF PROBLEM SOLVING METHODS

A. Dataset Formation

After analyzing and studying many sources about factors risks and reasons of cardiovascular-diseases, one of them is cardiologist's initial examination sheet, it was decided to collect new structure of dataset [2]. Dataset has data 224 patients and divided by 18 features. The feature of dataset is divided into 3. Objective, Anamnestic and Laboratory features. Objective feature – all people have these types of features, but the data of every person is different. Anamnestic features - a set of information obtained during a medical examination by questioning person. Laboratory features – a set of information

obtained using various specialized equipment in medical center [3].

Objective features:

Age: The age of person;

Gender: The person's gender (0 = women, 1 = man);

Height: The person's height in centimeters;

Weight: The person's weight in kilograms.

Anamnestic features:

Smoke: Does a person smoke (1 = yes, 0 = no);

Alcohol: Does a person drink alcohol (1 = yes, 0 = no);

Genetic predisposition: Did a close relative (grandfather, grandmother, father, mother) in the person's family get sick cardiovascular diseases (1 = yes, 0 = no);

Serious illness: Is person's has serious illnesses (1 = yes, 0 = no) [4].

Laboratory features:

Chest pain: 0: no angina, 1: typical angina, 2: atypical angina, 3: non-anginal pain.

RBP: The person's resting blood pressure (mm/Hg).

Total cholesterol: The person's total cholesterol (mg/dl) [5].

Fasting blood sugar: The person's fasting blood sugar (fbs > 120 mg/dl, 1 = true; 0 = false).

Resting ECG: (0 = normal, 1 = having ST-T wave abnormality, 2 = showing probable left ventricular hypertrophy by Estes' criteria).

MAHR: The maximum heart rate achieved by person.

Exercise-induced angina: (1 = yes; 0 = no).

ST depression: (1 = yes; 0 = no).

Thalassemia: A blood disorder (1 = yes; 0 = no).

B. Exploratory Data Analysis

The data used in this work are anonymous information about patients from clinical trials at the Nur-Avicenna medical center. There are data of 224 person, who has cardiovascular-diseases and who has not cardiovascular-diseases. The age of patients between 45 and 70, target audience is people who age is from 45 to 70. This target audience was chosen due to World health organization age classification, which consider people to 44 years as young. While mainly adults and old peoples suffer from CVD. Coronary heart disease develops earlier in men - from 40-45 years, and in women it is more common from 55-60 years.

To build a histogram by age, we use the `despot` function of the `seaborn` library. The code for plotting the histogram is shown in Fig.1.

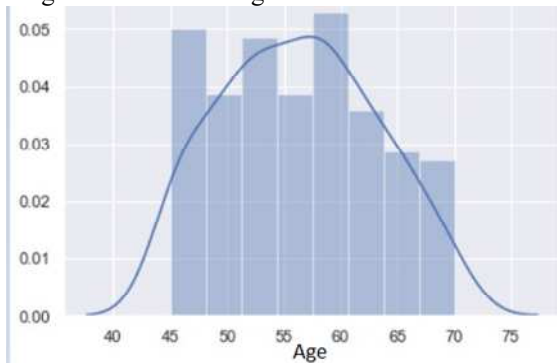


Figure 1. Histogram of distribution of people by age.

III. DEVELOPMENT OF MODELS

In the process of work, the several models were build using next algorithms [6]:

- Logistic Regression
- Random Forest
- Naïve Bayes
- SVM
- Decision Tree
- KNN

Logistic Regression (LR) is a powerful and well-established supervised classification technique.

It can be thought of as an extension of conventional regression, and it can only model a dichotomous variable that typically represents the occurrence or absence of an event. LR helps to find the probability that a new instance belongs to a certain class.

The support vector machine (SVM) algorithm can classify both linear and non-linear data. It first maps each data element into an n-dimensional feature space, where n is the number of features. It then defines a hyperplane that separates the data items into two classes, maximizing the marginal distance for both classes and minimizing classification errors.

Decision tree (DT) is one of the earliest and best-known machine learning algorithms. A decision tree models the decision logic, i.e., checks and matches the results to classify data items into a tree structure.

A random forest (RF) is an ensemble classifier that consists of a set of DTs, just as a forest is a collection of many trees. Very deep DTs often cause overfitting of the training data, resulting in a large scatter in classification results with little change in the input data.

Naive Bayes (NB) classification method based on Bayes' theorem. This theorem can describe the probability of an event based on a priori knowledge of the conditions associated with that event. This classifier assumes that a particular function in a class is not directly related to any other function, although the functions of that class may have interdependencies among themselves.

The K-nearest neighbor (KNN) algorithm is one of the simplest and first classification algorithms. It can be thought of as a simpler version of the NB classifier. Unlike the NB method, the KNN algorithm does not require probabilities to be considered.

All models were build using algorithms, which were imported from library `sklearn`.

The quality of these models was assessed using the previously described metrics based on test samples.

The sample was divided in the dataset into a "training" sample (75%) in which CVD risk algorithms and a "test" sample were obtained (the remaining 25%), which was used to test and evaluate algorithms. This is shown in fig.2.

```

from sklearn.model_selection import train_test_split

inputs = dataset.drop("output",axis=1)
output = dataset["output"]

X_train,X_test,Y_train,Y_test = train_test_split(inputs,output,test_size=0.25,random_state=42)

```

Figure 2. Train and test split.

The first model was built using Logistic Regression algorithm. The code for this machine learning algorithm is presented in Fig.3

```

from sklearn.linear_model import LogisticRegression

log_reg = LogisticRegression()

log_reg.fit(X_train,Y_train)

Y_pred_log_reg = log_reg.predict(X_test)

```

Figure 3. Logistic regression.

The result of metrics: accuracy, precision, recall for Logistic Regression is presented in Fig.4.

```

acc_score_lr = round(accuracy_score(Y_pred_log_reg,Y_test)*100,2)
pre_score_lr = round(precision_score(Y_pred_log_reg,Y_test)*100,2)
rec_score_lr = round(recall_score(Y_pred_log_reg,Y_test)*100,2)

print("The accuracy score achieved using Logistic Regression is: "+str(acc_score_lr)+" %")
print("The precision score achieved using Logistic Regression is: "+str(pre_score_lr)+" %")
print("The recall score achieved using Logistic Regression is: "+str(rec_score_lr)+" %")

```

```

The accuracy score achieved using Logistic Regression is: 87.5 %
The precision score achieved using Logistic Regression is: 90.62 %
The recall score achieved using Logistic Regression is: 87.88 %

```

Figure 4. Results of evaluation of metrics for logistic regression.

The second model was built using Support Vector Machine (SVM) algorithm [7]. The code for this machine learning algorithm is presented in Fig.5.

```

from sklearn import svm

sv = svm.SVC(kernel='linear')

sv.fit(X_train, Y_train)

Y_pred_svm = sv.predict(X_test)

```

Figure 5. Support vector machine.

The result of metrics: accuracy, precision, recall for Support Vector Machine is presented in Fig.6.

```

acc_score_svm = round(accuracy_score(Y_pred_svm,Y_test)*100,2)
pre_score_svm = round(precision_score(Y_pred_svm,Y_test)*100,2)
rec_score_svm = round(recall_score(Y_pred_svm,Y_test)*100,2)

print("The accuracy score achieved using SVM is: "+str(acc_score_svm)+" %")
print("The precision score achieved using SVM is: "+str(pre_score_svm)+" %")
print("The recall score achieved using SVM is: "+str(rec_score_lr)+" %")

The accuracy score achieved using SVM is: 78.57 %
The precision score achieved using SVM is: 81.25 %
The recall score achieved using SVM is: 87.88 %

```

Figure 6. Results of evaluation of metrics for support vector machine.

The third model was built using Naïve Bayes (NB) algorithm [8]. The code for this machine learning algorithm is presented in Fig.7.

```

from sklearn.naive_bayes import GaussianNB

nb = GaussianNB()

nb.fit(X_train,Y_train)

Y_pred_nb = nb.predict(X_test)

```

Figure 7. Naïve Bayes.

The result of metrics: accuracy, precision, recall for Naïve Bayes is presented in Fig.8.

```

acc_score_nb = round(accuracy_score(Y_pred_nb,Y_test)*100,2)
pre_score_nb = round(precision_score(Y_pred_nb,Y_test)*100,2)
rec_score_nb = round(recall_score(Y_pred_nb,Y_test)*100,2)

print("The accuracy score achieved using Naive Bayes is: "+str(acc_score_nb)+" %")
print("The precision score achieved using Naive Bayes is: "+str(pre_score_nb)+" %")
print("The recall score achieved using Naive Bayes is: "+str(rec_score_nb)+" %")

The accuracy score achieved using Naive Bayes is: 73.21 %
The precision score achieved using Naive Bayes is: 53.12 %
The recall score achieved using Naive Bayes is: 100.0 %

```

Figure 8. Results of evaluation of metrics for Naïve Bayes.

The result of metrics: accuracy, precision, recall for Random Forest is presented in Fig.9.

```

acc_score_randomfor = round(accuracy_score(Y_pred_randomfor,Y_test)*100,2)
pre_score_randomfor = round(precision_score(Y_pred_randomfor,Y_test)*100,2)
rec_score_randomfor = round(recall_score(Y_pred_randomfor,Y_test)*100,2)

print("The accuracy score achieved using Random Forest is: "+str(acc_score_randomfor)+" %")
print("The precision score achieved using Random Forest is: "+str(pre_score_randomfor)+" %")
print("The recall score achieved using Random Forest is: "+str(rec_score_randomfor)+" %")

The accuracy score achieved using Random Forest is: 94.64 %
The precision score achieved using Random Forest is: 93.75 %
The recall score achieved using Random Forest is: 96.77 %

```

Figure 9. Results of evaluation of metrics for Random Forest.

The sixth model was built using K Nearest Neighbors (KNN) algorithm [9]. The code for this machine learning algorithm is presented in Fig.10.

```
from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=9)
knn.fit(X_train,Y_train)
Y_pred_knn=knn.predict(X_test)
```

Figure 10. KNN.

The result of metrics: accuracy, precision, recall for KNN is presented in Fig.11.

```
acc_score_knn = round(accuracy_score(Y_pred_knn,Y_test)*100,2)
pre_score_knn = round(precision_score(Y_pred_knn,Y_test)*100,2)
rec_score_knn = round(recall_score(Y_pred_knn,Y_test)*100,2)

print("The accuracy score achieved using KNN is: "+str(acc_score_knn)+" %")
print("The precision score achieved using KNN is: "+str(pre_score_knn)+" %")
print("The recall score achieved using KNN is: "+str(rec_score_knn)+" %")

The accuracy score achieved using KNN is: 78.57 %
The precision score achieved using KNN is: 81.25 %
The recall score achieved using KNN is: 81.25 %
```

Figure 11. Results of evaluation of metrics for KNN.

After building models that predict the risks of CVD diseases, the quality of these models was assessed using the metrics already described based on test samples [10]. Metric values for the forecasting method are presented in the table 1 which is named result of metric for algorithms.

TABLE I. RESULT OF METRIC FOR ALGORITHMS

Algorithm	Accuracy	Precision	Recall
Logistic Regression	87.5%	90.62%	87.88%
Support Vector Machine	78.57%	81.25%	87.88%
Naïve Bayes	73.21%	53.12%	100.0%
Decision Tree	75.0%	68.75%	84.62%
Random Forest	94.64%	93.75%	96.77%
KNN	75.57%	81.25%	81.25%

As it shown the model built by using Random Forest algorithm shows the best result. It has 94.64% accuracy and 93.75% precision. Due to this, model learned by using Random Forest algorithm will be used in further in software for the diagnosis of cardiovascular diseases shown in Fig.12.

Figure 12. Software for the diagnosis of cardiovascular disease.

Software for the diagnosis of cardiovascular disease was created by program language Python. Mainly were used library PyQt5, which is designed to created Graphical User Interface. Firstly, the model is added to the program. Then, doctor should fill in all field, after that program gives result “Yes”, which means patient has problem with CVD or “No”, which means patient has no problem with CVD.

IV. DISCUSSION

The development of such a system helps to solve the problem with a shortage of medical workers. A district hospital doctor may not have sufficient expertise or experience to determine at an early-stage oncology or ischemic stroke, digital technologies will help him make the correct primary diagnosis and decide on the urgent referral of the patient to a specialized medical institution.

In paper, the current topic was considered - the usage of machine learning algorithms in medicine for the prediction and preventive diagnosis of diseases. Namely, the diagnosis of cardiovascular diseases (CVD). CVD was chosen because the leading cause of death worldwide: no other cause causes as many deaths each year as CVD. An estimated 17.9 million people died from CVD in 2016, accounting for 31% of all deaths worldwide. 85% of these deaths were due to heart attack and stroke.

This paper provides a definition and explanation of machine learning technologies. An overview of already implemented projects for the use of artificial intelligence in medicine in Kazakhstan and in the world is presented, such as Cerebra or IBM Watson [11].

The article shows the creation of 6 trained models for the diagnosis of cardiovascular diseases based on 6 different machine learning algorithms, such as Logistic regression, Support Vector Machine, Naïve Bayes, Decision tree, Random Forest, K Nearest Neighbors [12]. The quality of the model was evaluated by such metrics as the percentage of correct answers (accuracy), accuracy (precision) and completeness (recall) [13]. The best machine learning algorithm is determined. It is Random Forest algorithm, which gives about 95% of accuracy, 94 % of precision and 96 % of recall.

In the article was described software for the diagnosis of cardiovascular disease by using program language Python. For this software used model learned using by Random Forest.

V. CONCLUSION

It can be argued that new technologies make it possible to quickly analyze huge amounts of information on a specific task, based on which make predictions and issue recommendations to doctors. AI is especially in demand in the field of diagnostics this kind of innovation frees people from routine processes. A computer assistant, for example, can help a doctor analyze visual information: X-rays, ultrasound. It looks for patterns in medical images that are characteristic of diseases (tumors, hemorrhages, blockages of blood vessels) and highlights the smallest details that a person would not pay attention to. According to experts, such information makes up 90 of the total volume of all medical data, and its study takes a lot of time for a doctor.

At the same time, not the attendants, but the medical workers of the hospital face the most difficult mental tasks. Treating physicians must familiarize themselves with hundreds of different ways to offer medical support and advanced methods of examination, with an abundance of pharmaceuticals and up-to-date notes on their side effects. Specialists need to consider indications and contraindications, characteristics of patients, familiar cases of diseases. For effective work it is necessary to cultivate an avalanche of information. Automating all these tasks does indeed have the potential to save resources and increase the quality of proposals.

The development of a system aimed at automating some of the tasks of medical staff, mainly routine ones, can truly save resources and improve the quality of services. A doctor who has been bypassed from routine tasks will devote his time to a more intellectually challenging task.

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